

Problem 25.16

If $i = 0.04$, calculate the accumulated value of a continuous annuity payable at a rate of 100 per year for 10 years.

Solution

We are given the annual effective interest rate

$$i = 0.04.$$

The corresponding force of interest is

$$\delta = \ln(1 + i).$$

Substitute $i = 0.04$:

$$\delta = \ln(1.04).$$

The accumulated value of a continuous annuity payable at rate 100 per year for 10 years is

$$100\bar{s}_{\overline{10}|} = 100 \left(\frac{e^{10\delta} - 1}{\delta} \right).$$

Since $e^\delta = 1.04$, this becomes

$$100\bar{s}_{\overline{10}|} = 100 \left(\frac{(1.04)^{10} - 1}{\ln(1.04)} \right).$$

Thus,

$$AV = 1224.4659679.$$

Therefore, the accumulated value is

$$\boxed{1224.47}.$$